Unit 2 – Data Structures in R

Unit 2 provided a useful reintroduction to descriptive statistics, particularly measures of central tendency and dispersion. I found the Dong (2023) paper particularly helpful in contextualising these concepts and understanding both their practical applications and limitations. While descriptive statistics can summarise datasets and identify patterns, the paper highlighted that they cannot independently predict future outcomes, establish causal relationships or determine whether findings are statistically significant and generalisable to a wider population. I also found the discussion around the time required to collect and prepare data interesting, particularly when combining information from multiple sources. This reinforced my understanding that descriptive statistics represent an important starting point for data analysis, but often need to be supplemented by inferential techniques, such as regression analysis, to investigate more complex relationships.

Returning to mathematics after so many years has been something of a shock to the system, particularly as I have not studied some of these concepts in depth since my earlier university studies. However, I am gradually becoming more comfortable with the material and beginning to appreciate its relevance to the wider MSc programme.

The practical exercises also reminded me how frustrating coding in R can be, particularly when creating histograms and other visualisations. While generating the plots themselves was relatively straightforward, I spent considerably more time than anticipated adjusting their formatting, scales and appearance to produce something that looked more professional. Although frustrating at times, this was valuable practice in developing my R programming skills and understanding how to present statistical information more effectively.

References

Dong, Y. (2023) ‘Descriptive statistics and its applications’, Highlights in Science, Engineering and Technology, 47, pp. 16–23. Available at: https://doi.org/10.54097/hset.v47i.8159